# Supplementary material

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

This supplement contains the complete repeated-Euler comparisons and extended history-response diagnostics supporting the article. All tables read the same archived numerical data as the manuscript. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer; suffixes give nominal smooth-field order. GC denotes bounded global correction. BDF2 denotes the second-order backward differentiation formula.

## S1 Repeated Euler transfer comparisons

Table S1 includes the higher-order alternatives to the main-text comparison, and Table S2 reports all three realizations, including globally corrected NN1. The mesh-replacement interval is given in physical steps. The deformation parameter $s$ bounds streamwise displacement by the smallest incident corner-triangle altitude divided by $s$; $s = 2$ permits twice the displacement of $s = 4$. The 50% transition-position and 10–90% transition-width differences are measured against the fixed-mesh calculation in cells of spacing $h = 0.01$. With $Q_M(\rho) = \sum_i w_i^M \rho_i$, where $w_i^M$ is the median-dual volume, $\sum \Delta I_\rho$ is the signed sum of target-minus-source density-integral changes immediately across transfer, excluding physical evolution. Integral cancellation leaves local distortion unconstrained, so the tables report both integral and profile errors.

Table S1: Frequency comparison at safety factor $s = 4$ for all transfer variants. $\Delta n$ is the number of physical steps between mesh changes; $\Delta x/h$ and $\Delta w/h$ are signed position and width changes relative to the fixed-mesh reference.

| Transfer | $\Delta n$ | $\Delta x/h$ | $\Delta w/h$ | $\sum \Delta I_\rho$ |
|---|---|---|---|---|
| NN1 | 10 | 0.14 | 0.57 | $1.1 \cdot 10^{-4}$ |
| NN1 | 2 | 0.1 | 21.41 | $5 \cdot 10^{-4}$ |
| NN2 | 10 | −0.02 | −0.01 | $-8.8 \cdot 10^{-6}$ |
| NN2 | 2 | −0.04 | 0 | $-1.7 \cdot 10^{-5}$ |
| NN3 | 10 | 0.01 | −0.01 | $2.2 \cdot 10^{-5}$ |
| NN3 | 2 | 0.02 | 0 | $2.7 \cdot 10^{-5}$ |
| BAR2 | 10 | 0.01 | 0.25 | $2 \cdot 10^{-5}$ |
| BAR2 | 2 | −0.03 | 3.24 | $1.4 \cdot 10^{-4}$ |
| BAR3 | 10 | 0 | 0.01 | $-1.7 \cdot 10^{-5}$ |
| BAR3 | 2 | 0 | 0.07 | $-5.2 \cdot 10^{-6}$ |
| GP2 | 10 | 0.01 | 0 | $2.6 \cdot 10^{-6}$ |
| GP2 | 2 | 0.01 | 0 | $-2.8 \cdot 10^{-6}$ |
| GP3 | 10 | −0.01 | 0 | $-2.1 \cdot 10^{-5}$ |
| GP3 | 2 | −0.01 | 0.02 | $1.8 \cdot 10^{-5}$ |
| CV2 | 10 | 0 | 0 | $3 \cdot 10^{-14}$ |
| CV2 | 2 | 0 | 0.01 | $-8.2 \cdot 10^{-15}$ |

Table S2: High-frequency Euler comparison across three random deformation realizations. GC indicates whether the global correction is applied; $s = 2$ permits twice the displacement amplitude of $s = 4$. Position and width changes are in cells of the $h = 0.01$ reference mesh. The final column is the signed sum of target-minus-source density-integral changes at transfer.

| Transfer | GC | $s$ | Realization | $\Delta x/h$ | $\Delta w/h$ | $\sum \Delta I_\rho$ |
|---|---|---|---|---|---|---|
| BAR2 | No | 2 | 1 | $-0.06$ | 22.91 | $2.7 \cdot 10^{-4}$ |
| BAR2 | No | 2 | 2 | $-0.03$ | 22.98 | $3.1 \cdot 10^{-4}$ |
| BAR2 | No | 2 | 3 | $-0.07$ | 22.9 | $1.9 \cdot 10^{-4}$ |
| BAR2 | No | 4 | 1 | $-0.03$ | 3.24 | $1.4 \cdot 10^{-4}$ |
| BAR2 | No | 4 | 2 | 0 | 3.2 | $1.7 \cdot 10^{-4}$ |
| BAR2 | No | 4 | 3 | $-0.04$ | 3.12 | $6.6 \cdot 10^{-5}$ |
| CV2 | No | 2 | 1 | 0.19 | 0.06 | $-9.5 \cdot 10^{-15}$ |
| CV2 | No | 2 | 2 | 0.2 | 0.07 | $-4.2 \cdot 10^{-14}$ |
| CV2 | No | 2 | 3 | 0.18 | 0.05 | $7.9 \cdot 10^{-14}$ |
| CV2 | No | 4 | 1 | 0 | 0.01 | $-8.2 \cdot 10^{-15}$ |
| CV2 | No | 4 | 2 | 0.01 | $-0.01$ | $6.8 \cdot 10^{-14}$ |
| CV2 | No | 4 | 3 | 0 | 0.01 | $4.6 \cdot 10^{-14}$ |
| GP2 | No | 2 | 1 | 0.17 | 0.06 | $-1 \cdot 10^{-5}$ |
| GP2 | No | 2 | 2 | 0.19 | 0.06 | $-9.6 \cdot 10^{-6}$ |
| GP2 | No | 2 | 3 | 0.19 | 0.03 | $-9.5 \cdot 10^{-6}$ |
| GP2 | No | 4 | 1 | 0.01 | 0 | $-2.8 \cdot 10^{-6}$ |
| GP2 | No | 4 | 2 | 0.02 | 0 | $-3.1 \cdot 10^{-6}$ |
| GP2 | No | 4 | 3 | 0.01 | 0.01 | $-2.5 \cdot 10^{-6}$ |
| NN1 | Yes | 2 | 1 | $-0.91$ | 25.65 | $-2.8 \cdot 10^{-15}$ |
| NN1 | Yes | 2 | 2 | $-0.65$ | 23.8 | $7.9 \cdot 10^{-14}$ |
| NN1 | Yes | 2 | 3 | 0.01 | 26.33 | $1 \cdot 10^{-13}$ |
| NN1 | Yes | 4 | 1 | $-0.15$ | 18.43 | $-5.3 \cdot 10^{-14}$ |
| NN1 | Yes | 4 | 2 | $-0.01$ | 21.28 | $9.5 \cdot 10^{-15}$ |
| NN1 | Yes | 4 | 3 | $-0.11$ | 21.2 | $-1.9 \cdot 10^{-14}$ |
| NN1 | No | 2 | 1 | $-0.38$ | 26.36 | $1.3 \cdot 10^{-3}$ |
| NN1 | No | 2 | 2 | $-0.29$ | 24.98 | $-1.6 \cdot 10^{-3}$ |
| NN1 | No | 2 | 3 | 0.44 | 27.14 | $4.9 \cdot 10^{-4}$ |
| NN1 | No | 4 | 1 | 0.1 | 21.41 | $5 \cdot 10^{-4}$ |
| NN1 | No | 4 | 2 | 0.3 | 22.89 | $-8.9 \cdot 10^{-4}$ |
| NN1 | No | 4 | 3 | 0.2 | 22.75 | $8.3 \cdot 10^{-4}$ |

# S2 Boundary forcing and unequal-trial residuals

## S2.1 Slip-wall comparison

In the common-trial residual test, the predicted forcing is $-M_B\eta^n/\Delta t$, with $\eta^n = 2\varepsilon^n - \varepsilon^{n-1}/2$ formed from the latest and older target-referenced history discrepancies. Here $M_B$ is the receiving finite-volume mass matrix and $\Delta t$ the physical time step. The solver reports the negative of this residual. The wall-normal and tangential comparisons below concern the residual mismatch, not the transferred state error.

At a common trial state and the 370 slip-wall vertices, the wall-normal momentum mismatch is $3.4 \times 10^{-12}$ for GP2, $4.1 \times 10^{-12}$ for GP3 and CV2, and $4.5 \times 10^{-12}$ for NN1. The tangential mismatch is $(1.4\text{–}1.7) \times 10^{-11}$, a fraction $(1.1\text{–}1.3) \times 10^{-7}$ of the tangential history forcing of $1.25 \times 10^{-4}$. All four transfers return the same wall state, that of the source solution; its wall-normal momentum discrepancy from the native target state is $1.1 \times 10^{-14}$, so the normal history forcing is negligible and the slip-wall operation of the receiving equations does not alter it. These comparisons include the receiving slip-wall operation in both the predicted and measured forcing.

## S2.2 Residual probes in the cancellation experiment

In the cancellation experiment, the history discrepancies are $\varepsilon^n = e/4$ and $\varepsilon^{n-1} = e$, giving $\eta^n = 2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. The initial residual probes nevertheless use different trial states. Let $M_B$ be the receiving finite-volume mass matrix and $F_B$ its spatial residual. For a trial state $Y$ and displacement $q$, their difference has the form

$$\Delta\Phi = \frac{3M_Bq}{2\Delta t} + F_B(Y+q) - F_B(Y) - \frac{M_B\eta^n}{\Delta t}. \tag{S1}$$

With $q = e/4$ and $\eta^n = 0$, the temporal part is three quarters of the older-only forcing magnitude. The measured ratio near 0.75 also includes a spatial-residual difference and therefore does not verify the common-trial history-forcing identity. In contrast, at an identical trial state the history contribution to the residual difference is exactly $-M_B\eta^n/\Delta t$. Cancellation is temporary even for an exact first step: if the first-step state difference vanishes, the older level retains $e/4$, producing the subsequent history combination $-e/8$.

# S3 Perturbation-amplitude dependence

For the target-referenced state difference $\delta$, subscripts $f$, $n$, and $n-1$ denote equal spatial discrepancies imposed in both BDF2 histories, the latest only, and the older only. The superposition remainder is $\mathcal{E} = \delta_f - \delta_n - \delta_{n-1}$; its lift counterpart is $\mathcal{E}_L = \Delta C_{L,f} - \Delta C_{L,n} - \Delta C_{L,n-1}$. State norms are Euclidean or volume-weighted, $\|v\|_M = (v^T M_B v)^{1/2}$. Scaling the discrepancy by $\lambda = 1, 1/2, 1/4$ reduces the Euclidean-state and lift superposition remainders for BAR2, GP2, and CV2 (Table S3). The Euclidean absolute state remainder decreases by factors of about 3.2–3.5 per halving, faster than linearly over this range. The volume-weighted absolute remainder falls more slowly, from approximately $2.4 \times 10^{-5}$ to $1.5\text{–}1.7 \times 10^{-5}$, while its normalized value rises from about 3.4% to 8.7–9.2%. The directional response is diagnostic- and norm-dependent; these tests establish neither a uniform quadratic remainder nor a verified numerical floor in all norms.

Table S3: Amplitude dependence of first-step superposition. $\mathcal{E} = \delta_f - \delta_n - \delta_{n-1}$ is the state remainder and $\mathcal{E}_L$ its lift counterpart. Relative remainders use the corresponding both-histories response. The Euclidean and volume-weighted state metrics have different amplitude dependence.

| Transfer | $\lambda$ | $\|\mathcal{E}\|_2$ | $\|\mathcal{E}\|_2/\|\delta_f\|_2$ | $\|\mathcal{E}\|_M$ | $\|\mathcal{E}\|_M/\|\delta_f\|_M$ | $\|\mathcal{E}_L\|$ |
|---|---|---|---|---|---|---|
| BAR2 | 1 | $1.76 \cdot 10^{-3}$ | $7.78 \cdot 10^{-3}$ | $2.36 \cdot 10^{-5}$ | $3.36 \cdot 10^{-2}$ | $2.22 \cdot 10^{-5}$ |
| BAR2 | 0.5 | $5.45 \cdot 10^{-4}$ | $4.82 \cdot 10^{-3}$ | $1.97 \cdot 10^{-5}$ | $5.6 \cdot 10^{-2}$ | $4.74 \cdot 10^{-6}$ |
| BAR2 | 0.25 | $1.6 \cdot 10^{-4}$ | $2.83 \cdot 10^{-3}$ | $1.54 \cdot 10^{-5}$ | $8.74 \cdot 10^{-2}$ | $1.76 \cdot 10^{-6}$ |
| GP2 | 1 | $1.86 \cdot 10^{-3}$ | $8.23 \cdot 10^{-3}$ | $2.45 \cdot 10^{-5}$ | $3.44 \cdot 10^{-2}$ | $2.23 \cdot 10^{-5}$ |
| GP2 | 0.5 | $5.82 \cdot 10^{-4}$ | $5.15 \cdot 10^{-3}$ | $2.06 \cdot 10^{-5}$ | $5.77 \cdot 10^{-2}$ | $4.42 \cdot 10^{-6}$ |
| GP2 | 0.25 | $1.65 \cdot 10^{-4}$ | $2.93 \cdot 10^{-3}$ | $1.65 \cdot 10^{-5}$ | $9.24 \cdot 10^{-2}$ | $1.58 \cdot 10^{-6}$ |
| CV2 | 1 | $1.86 \cdot 10^{-3}$ | $8.22 \cdot 10^{-3}$ | $2.42 \cdot 10^{-5}$ | $3.4 \cdot 10^{-2}$ | $2.27 \cdot 10^{-5}$ |
| CV2 | 0.5 | $5.82 \cdot 10^{-4}$ | $5.15 \cdot 10^{-3}$ | $2.06 \cdot 10^{-5}$ | $5.77 \cdot 10^{-2}$ | $4.53 \cdot 10^{-6}$ |
| CV2 | 0.25 | $1.66 \cdot 10^{-4}$ | $2.95 \cdot 10^{-3}$ | $1.64 \cdot 10^{-5}$ | $9.16 \cdot 10^{-2}$ | $1.63 \cdot 10^{-6}$ |

## S4 Method-wise RANS inner convergence

The Spalart–Allmaras (SA) and shear-stress-transport (SST) calculations use BDF2 physical steps solved by inner iteration. The stopping target is $\log_{10} \mathrm{rms}(r_\rho) \leq -12$ in the solver density-residual normalization. Reaching the iteration limit alone does not establish convergence. Table S4 gives the full records behind the compact main-article comparison; terminal residuals are medians over physical steps, not residuals of averaged solutions. The nominal rows cover the full remeshing interval and other rows the shorter sensitivity interval; $\Delta t_0$ denotes the nominal physical time step.

Table S4: Inner convergence by method for the first deformation realization. Residuals are median terminal $\log_{10} \mathrm{rms}(r_\rho)$; $N_{\max}$ is the iteration limit.

| Closure | Transfer | $N_{\max}$ | $\Delta t/\Delta t_0$ | Physical steps | Steps at limit | Residual |
|---|---|---|---|---|---|---|
| SA | GP2 | 800 | 1 | 398 | 152 | -12.00 |
| SA | NN2 | 800 | 1 | 398 | 155 | -12.00 |
| SA | CV2 | 800 | 1 | 398 | 147 | -12.00 |
| SA | GP2 | 1600 | 1 | 38 | 0 | -12.00 |
| SA | NN2 | 1600 | 1 | 38 | 0 | -12.00 |
| SA | CV2 | 1600 | 1 | 38 | 0 | -12.00 |
| SA | GP2 | 800 | 1/2 | 78 | 0 | -12.00 |
| SA | NN2 | 800 | 1/2 | 78 | 0 | -12.00 |
| SA | CV2 | 800 | 1/2 | 78 | 0 | -12.00 |
| SA | GP2 | 800 | 1/4 | 158 | 0 | -12.00 |
| SA | NN2 | 800 | 1/4 | 158 | 0 | -12.00 |
| SA | CV2 | 800 | 1/4 | 158 | 0 | -12.00 |
| SST | GP2 | 800 | 1 | 398 | 333 | -11.10 |
| SST | NN2 | 800 | 1 | 398 | 324 | -11.24 |
| SST | CV2 | 800 | 1 | 398 | 336 | -11.11 |
| SST | GP2 | 1600 | 1 | 38 | 28 | -11.33 |
| SST | NN2 | 1600 | 1 | 38 | 28 | -11.33 |
| SST | CV2 | 1600 | 1 | 38 | 30 | -11.27 |
| SST | GP2 | 6400 | 1 | 38 | 28 | -11.27 |
| SST | NN2 | 6400 | 1 | 38 | 30 | -11.17 |
| SST | CV2 | 6400 | 1 | 38 | 29 | -11.18 |
| SST | GP2 | 800 | 1/2 | 78 | 56 | -11.27 |
| SST | NN2 | 800 | 1/2 | 78 | 58 | -11.32 |
| SST | CV2 | 800 | 1/2 | 78 | 60 | -11.23 |